

SP-08: Centered Rank-Two Sign Matrices

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Abstract

This continuation proves the conjectured rank-two candidate bound for every endpoint matrix whose centered sign matrix has rank at most two, in every dimension and throughout the normalized interval $-1 \leq a < 1$. An analytic monotonicity lemma retains integer block sizes, avoiding a fractional-mass relaxation. The negative-parameter comparison is analytic; the nonnegative comparison is certified by exact rational polynomial identities with an independent standard-library verifier. A consequence is the unrestricted strip $-1 \leq a \leq -1 + (1 + \sqrt{2})/n^2$, improving the coefficient $1/3$ in the supplied boundary-strip argument. **The unrestricted SP-08 conjecture is not proved here.** Higher-rank indefinite deficits remain outside the argument.

1 Problem, candidate, and status

Let $\mathcal{S}_n[a, 1]$ be the real symmetric $n \times n$ matrices all of whose entries, including diagonal entries, belong to $[a, 1]$. Here $n \geq 2$ and $-1 \leq a < 1$. For a symmetric matrix,

$$\text{spread}(A) = \lambda_{\max}(A) - \lambda_{\min}(A).$$

SP-08 asks whether an endpoint matrix of rank two always attains the maximum spread over this box $[1]$. Convexity gives an endpoint maximizer, but does not give its rank.

Put $d = 1 - a$. If z indicates a k -element subset, the matrix $B_k = J - dzz^T$, $1 \leq k < n$, has two nonzero eigenvalues of opposite signs. Its squared spread is

$$F(k) = n^2 + d\{2nk - (a + 3)k^2\}, \tag{1}$$

$$M_n(a)^2 = \max_{1 \leq k < n, k \in \mathbb{Z}} F(k). \tag{2}$$

Completing the square gives

$$F(t) = \frac{4n^2}{a + 3} - d(a + 3) \left(t - \frac{n}{a + 3} \right)^2. \tag{3}$$

An allowed integer nearest $n/(a + 3)$ is optimal in this family. In particular,

$$M_n(a)^2 \geq U := \frac{4n^2}{a + 3} - \frac{d(a + 3)}{4}. \tag{4}$$

The term subtracted in (4) is essential: the continuous optimum alone would not establish the finite-dimensional conjecture.

Theorem 1 (Centered sign rank at most two). *Set $c = (1 + a)/2$ and $r = (1 - a)/2$. If $A \in \mathcal{S}_n[a, 1]$ is an endpoint matrix and its sign matrix $S = (A - cJ)/r$ has rank at most two, then $\text{spread}(A) \leq M_n(a)$.*

Theorem 2 (Unrestricted boundary strip). *For every $n \geq 2$ and*

$$-1 \leq a \leq -1 + \frac{1 + \sqrt{2}}{n^2},$$

$B_{\lfloor n/2 \rfloor}$ maximizes spread over all of $\mathcal{S}_n[a, 1]$. There is no rank or inertia restriction in this statement.

Theorem 2 uses only the analytic, negative-parameter portion of Theorem 1; it does not depend on a numerical test or the nonnegative-parameter certificate.

2 Classification and compression

A symmetric rank-two sign matrix is indefinite and has both positive and negative diagonal entries. Indeed, if all diagonal signs were equal, every principal 2×2 minor would vanish; their sum, the product of the two nonzero eigenvalues, cannot vanish. Choose a positive and a negative diagonal position and switch one sign so that their principal matrix is

$$H_2 = \begin{pmatrix} 1 & 1 \\ 1 & -1 \end{pmatrix}, \quad H_2^{-1} = H_2/2.$$

The vanishing Schur complement classifies all columns by their two entries on these positions: they are $\pm(1, 1)^T$ or $\pm(1, -1)^T$. Consequently, up to permutation,

$$S = \Delta \begin{pmatrix} J_p & J_{p,q} \\ J_{q,p} & -J_q \end{pmatrix} \Delta, \quad p + q = n, \quad p, q \geq 1, \quad (5)$$

where Δ is diagonal with entries ± 1 . Let P and Q be the sums of these switching signs in the two groups. Thus $|P| \leq p$, $|Q| \leq q$, $P \equiv p \pmod{2}$, and $Q \equiv q \pmod{2}$. Define

$$R_0 = n - \frac{P^2}{p} - \frac{Q^2}{q}, \quad K = P^2 + 2PQ - Q^2.$$

The nonzero spectrum of $A = cJ + rS$ is contained in the spectrum of

$$T = r \begin{pmatrix} p & \sqrt{pq} & 0 \\ \sqrt{pq} & -q & 0 \\ 0 & 0 & 0 \end{pmatrix} + cvv^T, \quad v = \begin{pmatrix} P/\sqrt{p} \\ Q/\sqrt{q} \\ \sqrt{R_0} \end{pmatrix}. \quad (6)$$

A zero added in forming this compression can only increase the spread bound. We also use (6) as a continuous family for real P, Q in the indicated rectangle. This relaxation is used to move to a particular integer endpoint, not to discard the integer constraint.

For $c, r, R_0 > 0$ the eigenvalues have the form $\lambda \geq \eta > 0 > -\mu$. The characteristic polynomial is

$$\begin{aligned} f(z) &= z^3 - tz^2 + e_2z + D, \\ t &= cn + r(p - q), \\ e_2 &= -2r^2pq + cr\{n(p - q) - K\}, \quad D = 2cr^2pqR_0. \end{aligned} \quad (7)$$

Degenerate cases, including a possibly positive semidefinite rank-two limit, are handled by continuity of the three-dimensional spread.

3 An integer-preserving monotonicity reduction

At fixed R_0 , increasing K decreases e_2 and increases both extreme magnitudes:

$$\frac{\partial \lambda}{\partial K} = \frac{cr\lambda}{(\lambda + \mu)(\lambda - \eta)}, \quad \frac{\partial \mu}{\partial K} = \frac{cr\mu}{(\lambda + \mu)(\mu + \eta)}.$$

As e_2 decreases the extreme roots move outwards and the middle positive root moves downwards, so the relevant real-root branch does not collide. Replacing (P, Q) by $(|P|, |Q|)$ therefore cannot decrease spread. Assume below that $P, Q \geq 0$.

Lemma 3. *If $P + Q > 0$, then $\mu \geq rqP/(P + Q)$.*

Proof. Let γ be minus the smallest eigenvalue of the sign part compressed orthogonally to the all-ones vector. The cJ term vanishes there, so Rayleigh's principle gives $\mu \geq r\gamma$. For the continuous family (6), use $v^\perp$ in $\mathbb{R}^3$ in place of the all-ones orthogonal complement; $\|v\|^2 = n$, and the identical calculation applies. The two potentially nonzero eigenvalues of this compression have trace $p - q - K/n$ and product $-2pqR_0/n$. Thus γ is the positive root of

$$h(g) = g^2 + \left(p - q - \frac{K}{n}\right)g - \frac{2pqR_0}{n}.$$

Put $g_0 = qP/(P + Q)$, $x = p - P$, $y = q - Q$, and $N = P + Q$. Direct expansion gives $nN^2h(g_0) = -\mathcal{R}$, where

$$\begin{aligned} \mathcal{R} = & P^2NQ^2 + PQ(2P^2 + 5PQ + 4Q^2)x + PQ(4P^2 + 9PQ + 6Q^2)y \\ & + NQ(P + 2Q)x^2 + P(P^2 + 4PQ + 5Q^2)y^2 \\ & + 2(P^3 + 4P^2Q + 6PQ^2 + 2Q^3)xy \\ & + N(P + 2Q)x^2y + (P^2 + 4PQ + 2Q^2)xy^2 + PQy^3. \end{aligned}$$

Every term is nonnegative. Hence $h(g_0) \leq 0$, $\gamma \geq g_0$, and the claim follows. The identity is also checked by the supplied symbolic audit. $\square$

Lemma 4. *At fixed $p, q, Q \geq 0$, the spread in (6) is nondecreasing as P runs from 0 to p .*

Proof. Write $s = \lambda + \mu$ and

$$C = 2\lambda\mu + \eta(\lambda - \mu), \quad W = \lambda - \mu - 2\eta.$$

Differentiating (7) gives

$$\frac{ds}{dP} = \frac{2cr\{C(P + Q) - 2rqPW\}}{s(\lambda - \eta)(\mu + \eta)}. \quad (8)$$

The denominator is positive. Also $C > 0$: when $\lambda < \mu$, use $\eta \leq \lambda$ to see $C \geq \lambda\mu + \lambda\eta > 0$. If $W \leq 0$, the numerator is nonnegative. If $W > 0$, then $\lambda > \mu$, so Lemma 3 gives

$$C(P + Q) \geq 2\lambda\mu(P + Q) \geq 2\lambda rqP \geq 2rqPW.$$

The result follows; boundary points follow by continuity. $\square$

We can therefore set $P = p$ while leaving Q unchanged. Since Q retains its original parity and integrality, the numbers

$$y = \frac{q + Q}{2}, \quad z = \frac{q - Q}{2}$$

are integers. The resulting endpoint matrix has base

$$K_A(a) = \begin{pmatrix} 1 & 1 & a \\ 1 & a & 1 \\ a & 1 & a \end{pmatrix} \quad (9)$$

and multiplicities (p, y, z) . When $z = 0$ it is B_q . Otherwise $p, z \geq 1$ and $y \geq 1$. Its deficit has rank three when $p, y, z > 0$; it is not a case of the earlier rank-at-most-two deficit theorem.

4 A three-type comparison for negative parameters

For (9), let $q = y + z$, $n = p + q$, $b = 1 + a$, $d = 1 - a$, and $t = p + aq$. The characteristic coefficients are

$$e_2 = -dpq + dbz(p - q + z), \quad e_3 = -bd^2pyz.$$

Write its eigenvalues as $\lambda \geq \eta \geq 0 > -\mu$ and $s = \lambda + \mu$. An exact identity is

$$s^2 = F(q) - 4dbz(p - q + z) + 2t\eta - 3\eta^2. \quad (10)$$

Lemma 5. *For $-1 \leq a < 1$, the middle eigenvalue satisfies $\eta \leq dbz$.*

Proof. Set $h_0 = dbz$. The principal compression on the p and z groups is

$$R = \begin{pmatrix} p & a\sqrt{pz} \\ a\sqrt{pz} & az \end{pmatrix}, \quad g(w) = w^2 - (p + az)w + adpz.$$

Its smaller eigenvalue is at most h_0 . For $a \leq 0$ this follows from its nonpositive determinant. For $a \geq 0$, a Rayleigh quotient orthogonal to $(\sqrt{p}, \sqrt{z})^T$ gives $\lambda_{\min}(R) \leq dpz/(p + z) \leq dz \leq dbz$. Interlacing gives $\eta \leq \lambda_{\max}(R)$, while direct substitution gives

$$f(h_0) = h_0\{g(h_0) - db^2yz\}.$$

If $0 < h_0 < \eta$, then $g(h_0) \leq 0$ by its location between the eigenvalues of R . This contradicts $f(h_0) > 0$ between zero and the first positive root of f . Degenerate cases follow by continuity. $\square$

Suppose $-1 \leq a \leq 0$, so $0 \leq b \leq 1$. If $t \geq 0$, define $V = (1 - b)F(q) + bF(y)$. Equation (10) gives

$$V - s^2 = 2t(dbz - \eta) + d^2bz^2 + 3\eta^2 \geq 0.$$

Thus s^2 is bounded by a convex combination of candidate values. If $t < 0$, then $p < q$. With $\delta = q - p$, another exact identity gives

$$F(p) - s^2 = db \left\{ 4 \left(z - \frac{\delta}{2} \right)^2 + 2p\delta \right\} - 2t\eta + 3\eta^2 \geq 0.$$

Consequently

$$s^2 \leq \max\{F(p), F(q), F(y)\}, \quad -1 \leq a \leq 0. \quad (11)$$

For integer multiplicities these are candidate values. The harmless case $y = 0$ uses $F(0) = n^2 \leq M_n(a)^2$.

5 An exact nonnegative-parameter certificate

Theorem 6. *For real $p, z \geq 1$, $y \geq 0$, and $0 \leq a < 1$, the squared spread of the three-dimensional compression of (9) is at most*

$$U = \frac{4(p+y+z)^2}{a+3} - \frac{(1-a)(a+3)}{4}.$$

For integer multiplicities it is therefore at most $M_n(a)^2$.

5.1 The spectral criterion

Let t, e_2, e_3 be the characteristic coefficients of a real symmetric 3×3 matrix. Put

$$D_0 = t^2 - 3e_2,$$

$$\mathcal{D} = t^2 e_2^2 - 4e_2^3 - 4t^3 e_3 - 27e_3^2 + 18te_2 e_3.$$

The three squared pairwise eigenvalue differences are the roots of

$$q(u) = u(u - D_0)^2 - \mathcal{D}.$$

Their largest value s^2 is at least D_0 . Since $q'(u) = (u - D_0)(3u - D_0) \geq 0$ for $u \geq D_0$, it suffices to prove

$$U \geq D_0, \quad U(U - D_0)^2 \geq \mathcal{D}. \quad (12)$$

The relevant three-type coefficients are

$$t = p + a(y + z), \quad e_2 = (1 - a)\{-py + apz - (1 + a)yz\}, \quad e_3 = -(1 + a)(1 - a)^2 pyz.$$

5.2 Polynomial certificates and domain coverage

Define the polynomials

$$B = \frac{4(a+3)(U - D_0)}{1 - a}, \quad Q = \frac{64(a+3)^3\{U(U - D_0)^2 - \mathcal{D}\}}{(1 - a)^2}.$$

All prefactors have positive sign on the parameter domain. Introduce $a = T/(1 + T)$ with $T \geq 0$. The domain is covered by two charts:

Chart	p	y	z
Plus	$(a+2)v + u$	v	$1 + w$
Minus	$1 + v$	$(1 + v + u)/(a+2)$	$1 + w$

Here $u, v, w \geq 0$. The plus chart covers $p \geq (a+2)y$; the minus chart covers $p \leq (a+2)y$, $p \geq 1$. After clearing the recorded positive denominators, the plus-chart polynomials have positive coefficients. The minus-chart polynomials are explicit sums of binomial squares with nonnegative monomial multipliers and positive-coefficient remainders.

Every square entry $(\alpha, \beta, \gamma, k, w)$ represents

$$w\{4^k X^\alpha + 4^{-k} X^\beta - 2X^\gamma\}, \quad \alpha + \beta = 2\gamma, \quad w > 0.$$

If $m = \min(\alpha, \beta)$ coordinatewise, this is exactly

$$wX^m \left(2^k X^{(\alpha-m)/2} - 2^{-k} X^{(\beta-m)/2} \right)^2.$$

All half-exponents are nonnegative integers and all coefficients are rational. The positive-coefficient remainders are retained, not estimated numerically. This proves (12) and Theorem 6.

5.3 Independent reconstruction

The program `code/verify_typeA_certificate.py` uses only standard-library integer and rational arithmetic. It does not import the generator or optimizer. It independently reconstructs the spectral polynomials as follows. Let $L = 1 + T$, $R = 3 + 4T$, and express $p = p_N/m$, $y = y_N/m$, $z = z_N/m$ in either chart. Define

$$\begin{aligned} t_N &= Lp_N + T(y_N + z_N), \\ e_{2,N} &= -Lp_Ny_N + Tp_Nz_N - (L + T)y_Nz_N, \\ e_{3,N} &= -(L + T)p_Ny_Nz_N, \\ D_N &= t_N^2 - 3e_{2,N}, \\ U_N &= 16L^3(p_N + y_N + z_N)^2 - m^2R^2, \\ W_N &= U_N - 4RD_N. \end{aligned}$$

Let $\mathcal{D}_N$ be the cubic discriminant evaluated at these numerator coefficients. The independently reconstructed numerators are

$$Q_N = U_NW_N^2 - 64R^3\mathcal{D}_N, \quad B_N = W_N.$$

For the plus chart $m = L$, the certificate polynomials satisfy $Q_N = L^3Q_+$ and $B_N = LB_+$. For the minus chart $m = 2 + 3T$, they satisfy $Q_N = m^2Q_-$ and $B_N = mB_-$. The verifier checks these identities before checking every square and remainder coefficient. Thus it does not merely trust that a supplied polynomial is the required spectral expression.

Floating-point linear programming was used only in searching for squares. Its output was rationalized and independently checked exactly. The numerical sweeps in the archive are supplementary audits, not premises of a proof.

6 Completion of the centered-rank-two theorem

Lemmas 3–4 reduce rank-two sign matrices to (9) with integer multiplicities. Equation (11) handles negative parameters; Theorem 6 handles nonnegative ones. If S has rank one, write $S = \pm ss^T$ with $s_i \in \{-1, 1\}$. For the positive sign, $A = cJ + rss^T$ is positive semidefinite with trace n , so its spread is at most n . For the negative sign its squared two-dimensional spread is $n^2 - 4cr(\mathbf{1}^T s)^2 \leq n^2$; degenerate cases also have spread at most n . Since $F(1) > n^2$, neither case exceeds $M_n(a)$. This proves Theorem 1.

7 Proof of the improved unrestricted strip

Let S be a symmetric sign matrix with rank at least three. Choose a nonsingular 3×3 submatrix T . Such a sign matrix has singular values $2, 2, 1$. To see this, its determinant is a nonzero multiple of four and is bounded in absolute value by $\sqrt{27}$, so its absolute determinant is four. The Gram matrix has diagonal entries three and off-diagonal entries ± 1 ; its determinant forces eigenvalues $4, 4, 1$.

For any rank-at-most-two matrix X , the corresponding submatrix R has a unit null vector v . Hence

$$\|S - X\|_F^2 \geq \|T - R\|_F^2 \geq \|(T - R)v\|^2 = \|Tv\|^2 \geq 1.$$

Choose X to retain the two extreme eigenvalues of S . The middle spectral energy is at least one, whence

$$\text{spread}(S)^2 \leq 2(n^2 - 1).$$

Write $L = \sqrt{2n^2 - 2}$ and $A = cJ + rS$. Subadditivity gives $\text{spread}(A) \leq cn + rL$. For $\epsilon = n \bmod 2$ and $k = (n - \epsilon)/2$,

$$\text{spread}(B_k)^2 = n^2 + r^2(n - \epsilon)^2 + 2r\epsilon(n - \epsilon).$$

Put $C_\epsilon = (n - \epsilon)^2 - (L - n)^2$. An exact comparison is

$$\text{spread}(B_k)^2 - (cn + rL)^2 = r\{(2 - \epsilon) - cC_\epsilon\}. \quad (13)$$

For even n , write $\beta = L/n$. Then $C_0 = n^2\beta(2 - \beta) \leq n^2$. For odd n ,

$$C_1 = 2nL - 2n^2 - 2n + 3 < 2(\sqrt{2} - 1)n^2.$$

In both cases (13) is nonnegative if $c \leq (1 + \sqrt{2})/(2n^2)$. Matrices with sign rank at most two are already covered by Theorem 1. Within this strip,

$$0 \leq \frac{n}{2} - \frac{n}{2 + 2c} = \frac{nc}{2(1 + c)} \leq \frac{1 + \sqrt{2}}{4n} < \frac{1}{2},$$

so $\lfloor n/2 \rfloor$ is an allowed nearest integer in (3). Thus the balanced candidate attains $M_n(a)$. Convexity extends the endpoint comparison to every matrix in the entry box, proving Theorem 2.

For $n \geq 3$, the same calculation gives the sharper sufficient thresholds

$$c \leq \begin{cases} \frac{2}{2n\sqrt{2n^2 - 2} - 2n^2 + 2}, & n \text{ even}, \\ \frac{1}{2n\sqrt{2n^2 - 2} - 2n^2 - 2n + 3}, & n \text{ odd}. \end{cases}$$

They may be used with $a \leq 0$ so that the comparison remains entirely analytic.

8 Relation to the supplied work and remaining gap

The supplied earlier manuscripts treat continuous positive-semidefinite deficits $J - A \succeq 0$, binary deficits of rank at most two, and a narrower boundary strip. They are retained verbatim under **provenance**/, not silently rewritten as a complete solution. An inherited dimension-11 verification report is present, but the large certificate to which it refers was not supplied. That finite enumeration is not represented as independently reproduced by the present archive.

The new theorem concerns the rank of the *centered sign matrix* $S = (A - cJ)/r$, not the rank of the deficit $H = (J - A)/d$. These are distinct conditions. Combining the present result with the supplied low-deficit-rank results still leaves matrices for which both ranks are at least three and the deficit is indefinite.

The missing universal inequality is

$$\boxed{\text{spread}(J - (1 - a)H) \leq M_n(a) \quad \text{for every symmetric binary } H, \quad n \geq 2, \quad -1 \leq a < 1.}$$

No argument here forces an unrestricted maximizer into the covered classes. In particular, the rank-two Rayleigh difference $xx^T - yy^T$ does not force its entrywise sign matrix to have rank two. Nested-neighborhood merging is also not generally spread-increasing; the numerical exploration records counterexamples to that proposed reduction, not counterexamples to SP-08.

9 Reproducibility

The archive includes this manuscript and PDF, Markdown proofs, exact certificates, their independently checked rational identities, supplementary numerical tests, search scripts, and inherited source files. The file `STATUS.json` states the partial status explicitly. Start with

```
python code/verify_typeA_certificate.py
```

which has no third-party dependencies. The symbolic audits use SymPy; the numerical tests use NumPy. SciPy is needed only to rerun the certificate search, not to validate its output. The machine-readable reports distinguish exact verification from floating-point testing.

References

- [1] OpenProblemsInNLA contributors. *SP-08, eigenvalues and inverse problems*. Public problem statement: <https://github.com/ajt60gaibb/OpenProblemsInNLA/tree/main/eigenvalues-and-inverse-problems/SP-08>.
- [2] Supplied research archive, *SP08_partial_results_and_proofs.zip*, and supplied manuscripts *sp08.tex*, *boundary_strip.tex*, and *finite_certificates.tex*. Preserved in the archive's provenance directory. These are prior research materials, not an assertion that unrestricted SP-08 is solved.